Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-11

Number of questions: 15 **Total marks: 15 X 1= 15**

1. Which of the following is/are a challenge faced by IoT in healthcare?

a. Devices have limited computational capability
b. Devices have limited memory
c. Devices use low-power battery hence have short battery life
d. All of the above

Answer: (d)

Justification: Please refer to Slide 20 of Lecture 1 of Week 11

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2. What is the general action of healthcare body sensors in case some abnormalities are detected?

a. Sensors do nothing
b. Sensors automatically turned off
c. Sensors trigger an alarm
d. None of the above

Answer: (c)

Justification: Please refer to Slide 5 of Lecture 1 of Week 11

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3. Confidentiality, Integrity, _____ and Availability are the four security cornerstones of IIoT based Healthcare monitoring and management.

a. Accelerometer
b. Authentication
c. Automation
d. Auxiliary

Answer: (b)

Justification: Please refer to Slide 19 of Lecture 1 of Week 11

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4. Optimization and Autonomy are the features of which among the following IIoT models in the Power sector?.

a. SLA Model
b. Peripheral Model
c. Maturity Model
d. Component Model

Answer: (c)

Justification: Please refer to Slide 6 of Lecture 2 of Week 11

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5. In Smart grids supported by IIoT, data communication network is not integrated with the physical power transmission network?

a. True
b. False

Answer: (b)

Justification: Please refer to Slide 8 of Lecture 2 of Week 11.

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6. Consider the following two statements with respect to Passive RFID tags and Active RFID tags.

1. Passive RFID tags have no internal power source whereas active RFID has a battery power source
2. Passive RFID tags have longer range than Active RFID tags

a. Both Statements 1 and 2 are correct
b. Statement 1 is correct but 2 is not
c. Statement 1 is incorrect while 2 is correct
d. Both Statements 1 and 2 are incorrect

Answer: (b)

Justification: Please refer to Slides 10 and 11 of Lecture 3 of Week 11

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7. IIoT can be deployed in inventory management to automate which of the following activities?

a. Only Stock-In Activities
b. Only Stock-Out Activities
c. Both Stock-In and Stock-Out Activities

d. Neither Stock-In nor Stock-Out Activities

Answer: (c)

Justification: Please refer to Slide 4 of Lecture 3 of Week 11

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8. In the context of Network Security, what does behavioral analytics deal with?

a. Detects abnormal behavior by the network
b. Establishes a connection between the sender and receiver
c. Generates new routes across the network
d. None of the above

Answer: (a)

Justification: Please refer to Slide 13 of Lecture 4 of Week 11

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9. IP rights are a risk when using AR/VR?

a. True
b. False

Answer: (a)

Justification: Please refer to Slide 21 of Lecture 4 of Week 11

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10. Which of the following is NOT a stakeholder in Facility Management?

a. Buildings
b. Finance
c. People
d. Patents

Answer: (d)

Justification: Please refer to Slide 3 of Lecture 5 of Week 11

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11. Which of the following challenges arise due to ageing of inventory?

a. Increase in network traffic
b. Need for proactive, preventive and maintenance procedures
c. Recurring taxation levied by the Government
d. All of the above

Answer: (b)
Justification: Please refer to Slides 12 of Lecture 5 of Week 11

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12. In IIoT enabled facility management, logically where do subcontractors are located?

a. In between Owners and Customers
b. As a part of the owners
c. As a part of the customers
d. As a master authority of the owners

Answer: (a)

Justification: Please refer to Slide 5 of Lecture 5 of Week 11

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13. Optimized usage of which of the following resources is preferable?

a. Manpower
b. Assets
c. Technology
d. All of the above

Answer: (d)

Justification: Please refer to Slide 11 of Lecture 5 of Week 11

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14. Transponders and Beacons are types of what kind of RFID tags?

a. Active RFID tags
b. Passive RFID tags
c. Semi-passive tags
d. All of the above

Answer: (a)

Justification: Please refer to Page 297 of the book Introduction to Industrial Internet of Things and Industry 4.0 by S Misra, C Roy, A Mukherjee by CRC Press

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15. Real time and easy diagnosis are _____ of Healthcare IoT?

a. Advantages
b. Risks
c. Disadvantages
d. None of the above

Answer: (a)

Justification: Please refer to Page 298 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy by Cambridge University Press.